

CLINICAL RESEARCH PROJECT

Protocol #: 19-H-
Drug Names: Ibrutinib, Duvelisib
IND: exempt

Title: Duvelisib for Ibrutinib-Resistant Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma (CLL/SLL)

Short Title: Duvelisib for Ibrutinib-Resistant CLL

Keywords: Duvelisib, PI3K Inhibitor, Ibrutinib, BTK Inhibitor, CLL, SLL

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301-402-1806

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Subjects of Study:

Number

Sex

Age-range

33 evaluable

M/F

≥18

(Accrual Ceiling 36, to account for ineligible subjects)

PROTOCOL SUMMARY

SYNOPSIS

Background:

- In chronic lymphocytic leukemia (CLL) or small lymphocytic lymphoma (SLL), ibrutinib resistance is predominantly caused by somatic mutations in *BTK* and *PLCG2*.¹⁻³ Virtually all patients with detectable mutations eventually develop progressive disease.^{2,4} Patients who discontinue ibrutinib often have rapidly progressive disease that can be difficult to control.⁵ These observations suggest that BTK inhibitors still exert at least a partial anti-tumor effect. Outcomes after ibrutinib discontinuation are poor.² Early detection of *BTK* and *PLCG2* mutations represents an opportunity for preemptive intervention to eliminate the resistant clone.²
- Duvelisib is a dual PI3K- γ and δ inhibitor. Duvelisib monotherapy improved progression-free survival and overall response rate compared to ofatumumab and had a manageable safety profile in subjects with previously treated CLL/SLL.⁶ Based on these results, duvelisib received US approval for CLL/SLL after at least 2 prior therapies.
- This study will assess duvelisib in patients who develop disease progression or *BTK* and/or *PLCG2* mutations on ibrutinib. Duvelisib will overlap with ibrutinib for the first six 28-day cycles to prevent disease acceleration often seen in patients who discontinue ibrutinib.

Primary Objective:

- To investigate the rate of overall response to duvelisib in patients with ibrutinib-resistant CLL.

Key Eligibility Criteria:

- Patients on current treatment for CLL/SLL with ibrutinib and at least one of the following:
 - *BTK* and/or *PLCG2* mutations
 - Progressive CLL per iwCLL guidelines⁷
- Patients with known Richter transformation will be excluded.

Design:

- This is a single-center, single-arm, open-label phase 2 study with a safety lead-in cohort.
- Treatment plan: Duvelisib will be administered with ibrutinib for the first six 28-day cycles then duvelisib monotherapy will be administered continuously until disease progression or intolerance.

Study Duration: 5 years.

Participant Duration: until disease progression or intolerance.

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STATEMENT OF COMPLIANCE

The trial will be carried out in accordance with International Conference on Harmonisation Good Clinical Practice (ICH GCP) and the following:

- United States (US) Code of Federal Regulations (CFR) applicable to clinical studies (45 CFR Part 46, 21 CFR Part 50, 21 CFR Part 56, 21 CFR Part 312, and/or 21 CFR Part 812)

National Institutes of Health (NIH)-funded investigators and clinical trial site staff who are responsible for the conduct, management, or oversight of NIH-funded clinical trials have completed Human Subjects Protection and ICH GCP Training.

The protocol, informed consent form(s), recruitment materials, and all participant materials will be submitted to the Institutional Review Board (IRB) for review and approval. Approval of both the protocol and the consent form must be obtained before any participant is enrolled. Any amendment to the protocol will require review and approval by the IRB before the changes are implemented to the study. In addition, all changes to the consent form will be IRB-approved; a determination will be made regarding whether a new consent needs to be obtained from participants who provided consent, using a previously approved consent form.

1.0 OBJECTIVES

1.1 Primary Objective

- To investigate the rate of overall response to duvelisib in patients with ibrutinib-resistant CLL

1.2 Secondary Objectives

- Progression-free survival (time from treatment initiation to progression of disease or death from any cause)
- Overall survival (time from treatment initiation to death from any cause)
- Duration of response (time from initial response to progression of disease)
- Best response
- Safety of duvelisib plus ibrutinib combination
- Safety of duvelisib monotherapy

1.3 Exploratory Objectives

- Immune cell profiling at baseline, during combination treatment with ibrutinib plus duvelisib, and during duvelisib monotherapy
- Minimal residual disease by flow cytometry in peripheral blood and/or bone marrow
- Quantification of allele frequency of *BTK* and *PLCG2* mutations (if present at baseline) during treatment
- Targeted or whole exome sequencing of sequential tumor samples to assess clonal evolution.

2.0 BACKGROUND & RATIONALE

2.1 Chronic Lymphocytic Leukemia and/or Small Lymphocytic Lymphoma

The World Health Organization classification of hematopoietic neoplasias describes CLL as a leukemic, lymphocytic lymphoma, being only distinguishable from SLL by its leukemic appearance.⁸ In the National Cancer Institute (NCI)-Working Group (WG) guidelines, the diagnosis of CLL requires the presence of at least 5×10^9 clonal B lymphocytes/L ($5000/\mu\text{L}$) in the peripheral blood. The definition of SLL requires the presence of lymphadenopathy and/or splenomegaly. Clonal B lymphocytes in the peripheral blood should not exceed $5 \times 10^9/\text{L}$ and the diagnosis should be confirmed by histopathologic evaluation of a lymph node biopsy when possible.⁹

Chronic lymphocytic leukemia/small lymphocytic lymphoma cells co-express the T-cell antigen CD5 and B-cell surface antigens CD19, CD20, and CD23. The levels of surface immunoglobulin, CD20, and CD79b are characteristically low compared with those found on normal B cells. Each clone of leukemia cells is restricted to expression of either kappa or lambda immunoglobulin light chains.⁹

For the purposes of this study, the term “CLL” will encompass both CLL and SLL.

2.2 Pathophysiology

CLL is characterized by a progressive accumulation of functionally incompetent lymphocytes. It is believed that antigenic stimulation, along with interactions of accessory cells, cytokines, and tumor microenvironment are promoting factors that stimulate proliferation of CLL cells and allows them to avoid apoptosis. These effects may differ in distinct CLL subgroups and thereby lead to the heterogeneity seen in clinical outcomes among individual cases.

The tumor microenvironment is known to play an important role in the development and maintenance of CLL. Important contributors to this permissive microenvironment include cellular components such as macrophages, T cells, or stromal follicular dendritic cells, and essential proteins (chemokines, cytokines, and angiogenic factors) which exert their oncogenic support via interactions with membrane receptors on the malignant cells. CLL cultured *in vitro* undergo apoptosis, but they can be rescued by coculturing with stromal cells or in presence of soluble factors.¹⁰

In human, T and leukemic B cells aggregate to form ‘proliferation centers’ within bone marrow (BM) and lymphoid tissues, where these rallied cells form immune synapses.¹¹ CD40 ligand form a crosstalk between T and CLL cells which stimulates leukemic B cell survival and promotes resistance to apoptosis.^{12,13} In addition, T cells in CLL are functionally¹¹ and phenotypically¹⁴ defective, and allow malignant B cells to escape immune surveillance. “Pseudo-exhaustion” of effector T cells is explained by increased expression of immune checkpoints on CLL cells^{11,15} and by increased number of regulatory T cells in CLL patients.¹⁶

Myeloid cells, in particular tumor associated macrophages (TAMs) and myeloid derived suppressor cells (MDSCs) are also important in the pathogenesis of CLL.^{17,18} *In vitro*, TAMs support the survival of CLL cells by cell-cell contact and the secretion of immunosuppressive cytokines.^{19,20} MDSCs are immunosuppressive cells that are found at increased frequency in CLL patients compared to healthy individuals and are associated with shorter lymphocyte doubling time.²¹⁻²³ Both TAMs and MDSCs impair tumor immune surveillance via expansion of Treg cells and suppression of T-cell activation in CLL.^{17,21,24}

2.3 Epidemiology and Clinical Course of Chronic Lymphocytic Leukemia

Chronic lymphocytic leukemia is the most common leukemia of adults in Western countries with an annual incidence of 2 to 4.5 per 100,000 in the general population. According to the NCI Surveillance Epidemiology and End Results (SEER) data, it is estimated that 14,990 men and women (8,870 men and 6,120 women) will be diagnosed with CLL. It affects males twice as frequently as females and is a disease of older individuals (median age 73.0 years for Whites).²⁵ The highest rates of incidence are seen in Caucasians followed by African Americans. Lower rates of incidence are seen in Asian and Hispanic populations.

The complications of CLL usually arise from progressive disease. These complications are, but not limited to infection, hematologic abnormalities, immune phenomena, secondary cancers, and disease transformation. It has been estimated that up to 70% of patients with CLL will develop infections, and infections account for about 55% of deaths in patients with CLL. Hematologic abnormalities include autoimmune hemolytic anemia (AIHA) in approximately 37% of patients, pure red cell aplasia in 6% of patients, and immune mediated thrombocytopenia in 2 to 4% of patients.²⁶ Secondary cancers are noted in about 11% of cases such as acute myeloid leukemia and other solid malignancies. Finally, disease transformation into other forms or more aggressive lymphomas such as diffuse large B-cell lymphoma and Hodgkin’s lymphoma occur in about 5% of cases.²⁷

2.4 Treatment Options for Ibrutinib-Resistant CLL/SLL

Patients who develop CLL progression on ibrutinib often have limited treatment options. Salvage therapies with reported outcomes are described below.

2.4.1 Salvage Chemoimmunotherapy

In a retrospective cohort study of 391 CLL patients treated with frontline ibrutinib, 55 patient were treated with a subsequent therapy after ibrutinib discontinuation.²⁸ Anti-CD20 monoclonal antibody-based chemoimmunotherapy produced an ORR of 50% among 9 patients. However, only 1 had discontinued ibrutinib for CLL progression. Further, a major limitation of conventional chemoimmunotherapy is toxicity. Rates of treatment-related mortality are 3.9% with fludarabine, cyclophosphamide, and rituximab (FCR) and 2.1% with bendamustine and rituximab (BR).²⁹ More than 90% of patients treated with FCR and 78.5% treated with BR experience moderate to severe adverse events equivalent to CTCAE Grade 3 to 5.²⁹ Hematologic toxicities such